

AIMO Proof Pilot — Final Grade

Problem (#_ID): 2_ALTORO

Submission: model_8

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one): 6

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

An essentially complete solution. It defines $X = EF \cap BC$ as the radical centre of the relevant circles, deduces $Q \in (AEHF)$ so $\angle AQH = 90^\circ$, identifies P as the midpoint of HQ via the nine-point homothety, and computes the required lengths to the correct answer. Several steps carry small justification gaps: X is used as the harmonic conjugate of D with respect to B, C (giving $XD = 140/3$) without proof, the height $35/12$ of H is not derived, and the Ptolemy relation for EQ is stated with a missing intermediate step. Each is locally repairable and individually minor — CG independently reaches the same 6 via the unproved $(X, D; B, C) = -1$. Essentially complete with a minor error, so scores **6**.